

Bending and durability performance of TRC permanent formwork under chloride corrosion and impressed current

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ABSTRACT

In order to further promote the application prospects of Textile Reinforced Concrete (TRC) permanent formwork, this paper used four-point bending test to investigate the mechanical properties of specimen under the action of chloride corrosion and impressed current with different chloride concentration, quantity of electric charge density and textile layers. Research has shown that compared to specimens in conventional environments and single chloride corrosion, when the specimens are subjected to chloride corrosion and impressed current, the failure mode of the specimens changes from balanced-reinforced failure to under-reinforced failure. The number of cracks on the outer surface of the specimens is significantly reduced, and the ultimate mid span deflection and load of the specimens are significantly reduced. When the chloride concentration remains constant, the bending performance of the specimen gradually decreases with the increase of electric charge density. In addition, increasing the number of textile layers in the specimen can significantly improve the bending performance of the specimen under chloride corrosion and impressed current. Finally, based on a series of basic assumptions and the analysis of the cross-sectional stress of the specimen when it was damaged, and combined with the experimental data in this paper, a calculation model that can effectively predict the four-point bending capacity of the specimen under different chloride concentrations and quantity of electric charge densities was obtained.

1. Introduction

At present, the green and low-carbon transformation of the construction industry is crucial to achieving carbon neutrality, which can be achieved through the promotion of renewable resources, recycling of waste materials [1] and low-carbon building materials [2]. In addition, for the improvement of the mechanical properties of concrete structures, it can be divided into aspects such as concrete-encased steel columns [3, 4], permanent formwork composite structures [5–13], etc. This paper mainly studies permanent formwork, which is a formwork that does not need to be removed during construction and is usually used as surface protection or support for concrete components. This type of formwork can save the consumption of traditional formwork materials, extend the life of the building and improve the energy efficiency of the building. It

is one of the important ways to realize the concept of green building. Therefore, many scholars have conducted extensive research on permanent formwork, proposed steel mesh cement "U" permanent formwork [5], pseudo-ductile cementitious composites (PDCC) permanent formwork [6], ultra high toughness cementitious composites (UHTCC) permanent formwork [7], fiber reinforced polymer (FRP) permanent formwork [8–10] and TRC permanent formwork [11–13].

In addition, existing research reports indicate that the corrosion rate of steel bars in reinforced concrete buildings located in tidal areas is significantly higher than that in seawater flooded areas and atmospheric areas. This is because building structures in tidal areas not only experience the intrusion of chloride, but also have sufficient oxygen, and the corrosion process requires the presence of oxygen. The coexistence of chloride and oxygen causes the corrosion reaction of steel bars to

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continue to increase [14,15]. At present, in order to extend the whole life cycle of the reinforced concrete building, the solution of the steel bar corrosion problem in the reinforced concrete building structure has become a hot issue for many researchers to think about, and many solutions have been put forward from the protection of the steel bar and concrete. In practical engineering, the commonly used steel bar corrosion protection methods in construction engineering mainly include the following methods, concrete local repair method, concrete coating method, steel bar coating method [16] and impressed current cathodic protection system [17,18].

TRC is a new high-performance cement based composite material, is a new composite material composed of carbon textile and conductive fine-grained concrete. It has the characteristics of conductivity, high strength, corrosion resistance and strong plasticity [19,20]. It can be used as anode material for impressed current cathodic protection system. In addition, the combination of TRC and ICCP can be used to prepare concrete from sea sand in coastal areas, which can effectively solve the problems of inconvenient transportation of raw materials [21] and shortage of river sand resources [22]. Currently, many scholars have conducted different studies on the bending performance of TRC plates in different environments. In conventional environments, Zargaran et al. [23] found that reducing the mesh size of textile can improve the ultimate bearing capacity of TRC plates. Halvaei et al. [24] found that impregnating textile with epoxy resin can significantly improve the toughness of TRC plates. In addition, Peled [25] pointed out that applying prestress to textile can improve the mechanical properties of TRC plates, but the material properties of textile are crucial, and textile with high elastic modulus should be selected to apply prestress. Xu et al. [26] and Pakravan et al. [27] found that impregnating textile with adhesive sand had little effect on cracking load, but could improve the crack development morphology. At the same time, the addition of chopped fibers can improve the cracking load and ultimate bearing capacity of TRC plates, effectively controlling the development of cracks. In addition, some scholars have further studied the bending performance of TRC plates in harsh environments based on their exploration of the bending performance of TRC plates in conventional environments. Du et al. [28] studied the bending performance of TRC plates under seawater corrosion by four-point bending test, which showed that the corrosion resistance of TRC plates is better, and the application of prestressing force can enhance the corrosion resistance; the deterioration of TRC plates will be accelerated when the coupling of both loading and corrosion environment. Yin et al. [29,30] investigated the mechanical properties of TRC plates under chloride dry and wet cycling and freeze-thaw cycling by means of four-point bending tests, and found that with the increase of the number of dry and wet cycling, the interfacial bond strength between the textile and the fine-grained concrete gradually decreases, and with the increase of the number of freeze-thaw cycling, the flexural load bearing capacity decreases. In addition, Hou et al. [31] explored the mechanical properties of TRC plates after high temperature by four-point bending test, and found that the flexural capacity of TRC plates was decreasing with the increase of temperature, mainly because of the combined effect of concrete matrix damage, textile deterioration, and the deterioration of the interfacial bond between the two at high temperatures.

In summary, the TRC permanent formwork in the harsh environment shows good bending performance and corrosion resistance, which lays a solid theoretical foundation for the construction of a new structural system using the TRC permanent formwork as an anode material combined with the impressed current cathodic protection method in the chloride environment. However, further clarification is still needed on the flexural performance of TRC permanent formwork under the dual effects of chloride environment and impressed current. In this paper, the damage mechanism of TRC permanent formwork under different chloride concentrations, quantity of electric charge densities and textile layers is investigated through four-point bending test, and the related four-point bending bearing capacity calculation model is established.

2. Experimental program

2.1. Conductive fine-grained concrete

The mix ratio of conductive fine-grained concrete is shown in Table 1. Among them, the basic parameters of chopped carbon fiber are shown in Table 2. The basic parameters of graphite are shown in Table 3. The basic parameters of hydroxypropyl methyl cellulose are shown in Table 4. In order to evenly disperse carbon fibers in fine-grained concrete, 200,000 viscosity hydroxypropyl methyl cellulose is selected as the carbon fiber dispersant, with a dosage of 20 % of the carbon fiber mass. The pouring, conductivity, and mechanical properties of conductive fine-grained concrete are detailed in references [32,33].

2.2. Textile

As shown in Fig. 1, the composition of the textile mainly consists of mutually perpendicular carbon fiber bundles and glass fiber bundles, in which the carbon fibers as reinforcing fibers are mainly applied in the stress direction, and the glass fibers as fixing fibers are mainly applied in the non-stress direction. The mechanical properties of the carbon fiber bundles are shown in Table 5.

2.3. Simulated seawater solution

In order to truly simulate the damage caused by long-term seawater solution environment to the mechanical properties of the TRC permanent formwork, this paper simulates seawater according to the various ion contents in seawater specified by ASTM [34]. The chemical composition of simulated seawater is shown in Table 6.

2.4. Specimen design and production

2.4.1. Specimen design

In order to explore the bending performance of the TRC permanent formwork under the action of chloride corrosion and impressed current, a four-point bending test was used to explore the damage and degradation mechanism of the TRC permanent formwork under different chloride concentrations, quantity of electric charge density and textile layers. Among them, three chloride solutions with different concentrations were set up with the normal seawater chloride concentration of 2 % as the minimum value to comprehensively evaluate its performance under various chloride concentrations. In addition, based on the principle of equal charge, applies a higher current density to the TRC permanent formwork in the experiment, thereby simulating its long-term performance under standard current density in a shorter time [35]. The grouping of specimens under four-point bending is shown in Table 7.

2.4.2. Specimen production

In order to reduce the discreteness of the TRC permanent formwork production process, the complexity of construction and improve the efficiency of specimen pouring, the large-size permanent formwork was poured first, and then use an electric cutting machine to cut the main-tained large-size permanent formwork into the size required for the four-point bending test (400 mm × 100 mm × 12 mm).

For the production process of TRC permanent formwork, the specific steps are as follows: (1) Cut the textile according to the size required by the TRC permanent formwork; (2) Cut the wooden board according to the size required by the TRC permanent formwork to make a bottom mold, and apply machine oil in the bottom mold to facilitate later demoulding; (3) Use an air nail gun to arrange 6 mm thick wooden strips around the bottom mold, pour the mixed conductive fine-grained concrete evenly on the bottom form and compact it; (4) After impregnating the textile with cement slurry, lay it straight and flat on the conductive fine-grained concrete. The carbon fiber in the textile are arranged in the direction of stress. Press the same wooden strips on the wooden strips

Table 1
Conductive fine-grained concrete (kg/m³).

Cement /42.5 R	Coarse sand /0.6–1.2 mm	Fine sand /0–0.6 mm	Flyash	Silica fume	Water	Water reducer	Carbon fiber /Mass fraction of cement	Graphite /Mass fraction of cement	Hydroxy propyl methyl cellulose /Mass fraction of carbon fiber
475	920	460	168	35	262	9.1	0.25 %	10.0 %	20.0 %

Table 2
Basic parameters of chopped carbon fiber.

Material	Tensile strength /MPa	Tensile modulus /GPa	Density /g/cm ³	Resistivity /Ω·m	Length /mm
T700	4900	230	1.75	1.6×10^{-5}	6

Table 3
Basic parameters of graphite.

Material	Mesh number	Carbon content /%	Density /g/cm ³
Graphite	200	90–98	1.9

Table 4
Basic parameters of hydroxypropyl methyl cellulose.

Material	Mesh number	Apparent density /g/cm ³	Carbonization temperature /°C	Discolor temperature /°C
HPMC	100	0.5	280–300	190–200

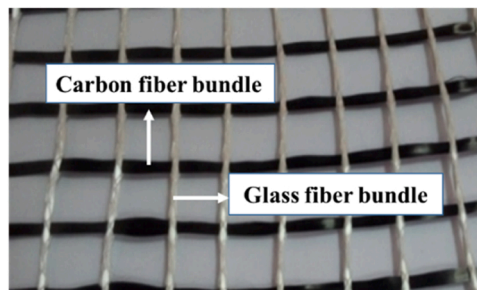


Fig. 1. Textile.

Table 5
Mechanical properties of carbon fiber bundles.

Fiber type	Elasticity Modulus /GPa	Ultimate strain /%	Tensile strength /MPa	Area /mm ²	Density /g/cm ³
Carbon fiber bundles	179.9	2.00	3600	0.45	1.8

described in step (3), press the textile on the first layer of fine-grained concrete, continue pouring fine-grained concrete in the second layer of wooden strips, and plaster flat; (5) For specimens containing two

Table 6
Chemical composition of simulated seawater(g/L).

Chloride concentration	NaCl	MgCl ₂	Na ₂ SO ₄	CaCl ₂	KCl	NaHCO ₃	KBr	H ₃ BO ₃	SrCl ₂	NaF
2 %	24.53	5.20	4.09	1.16	0.695	0.201	0.101	0.027	0.025	0.003
4 %	59.43	5.20	4.09	1.16	0.695	0.201	0.101	0.027	0.025	0.003
6 %	93.23	5.20	4.09	1.16	0.695	0.201	0.101	0.027	0.025	0.003

layers of textile, repeat the above steps; (6) After curing the TRC permanent formwork for 24 hours, remove the wooden strips and then maintain it in a standard curing room for 28 days.

2.4.3. Specimen corrosion

This paper uses the simulated seawater mix ratio described in Table 6 as the pool solution. As shown in Fig. 2(a), the two materials form a complete circuit through the solution and wires, in which the permanent formwork of the anode material is connected to the anode of the external DC power supply, and the stainless steel plate is connected to the negative pole of the external DC power supply. As shown in Fig. 2(b), in order for the TRC permanent formwork to act as an anode in the entire system, power is supplied through a DC power box, and constant voltage and constant current equipment is used to control the precise current endured by different specimens. In the pool, the TRC permanent formwork is placed on one side and the cathode material stainless steel plate is placed on the other side. The two materials form a complete circuit through the solution and wires, and are 50 mm apart.

2.5. Test loading scheme and measuring point layout

As shown in Fig. 3, the specimen was loaded using an electronic universal testing machine. The design dimensions of the specimen were designed with reference to [36]. The length of the specimen was 400 mm, the width was 100 mm, the span of the two supports was 300 mm, and the specimen loading rate was 0.5 mm/min. As shown in Fig. 4, this paper uses a special four-point bending loading device, which is equipped with a ball joint in the middle, so that the force on the specimen in the four-point bending test can be directed vertically downward, and the upper part of the loading end can form a whole with the load sensor through screws. In addition, during the test, a thin strip of steel plate was placed in the middle of the specimen, and a displacement sensor was placed at both ends of the steel plate to measure the mid-span deflection. Load sensors and displacement sensors collect data through equipment, including first crack load, first crack mid-span deflection, ultimate load, and ultimate mid-span deflection. The specific loading details of the specimen are shown in Fig. 5.

3. Test results and analysis

3.1. Test results

According to the test results, the first crack mid-span deflection, first crack load, ultimate mid-span deflection, ultimate load and failure mode of the TRC formwork under four-point bending are summarized as follows. See Table 8 for details.

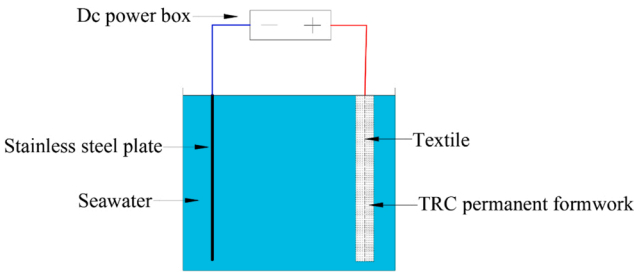
3.2. Appearance changes of specimen during impressed current

As shown in Fig. 6, when the specimen is placed in the chloride

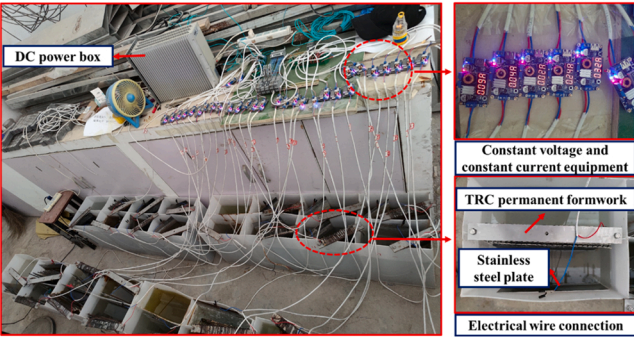
Table 7
Specimen Grouping.

Specimen	$A_a \times 10^{-3} / \text{m}^2$	$i / \text{mA} / \text{m}^2$	I / mA	Time / days	Chloride concentration / %	$q_a / 10^7 \text{C} / \text{m}^2$	Textile layers	Number
B0	42.72	0	0	-	0	0	1	3
B1	42.72	0	0	-	2	0	1	3
B2	42.72	0	0	-	4	0	1	3
B3	42.72	0	0	-	6	0	1	3
B4	42.72	540	23.07	75	2	0.35	1	3
B5	42.72	540	23.07	75	4	0.35	1	3
B6	42.72	540	23.07	75	6	0.35	1	3
B7	42.72	1080	46.14	75	2	0.7	1	3
B8	42.72	2160	92.28	75	2	1.4	1	3
B9	42.72	5400	230.688	75	2	3.5	1	3
B10	42.72	540	23.07	75	2	0.35	2	3

Note: A_a : Anode working area; i : Current density; I : Anode current value; q_a : Quantity of electric charge densities



(a) TRC permanent template power-on circuit diagram



(b) TRC permanent formwork power-on field diagram

Fig. 2. TRC permanent formwork under the action of chloride corrosion and impressed current.

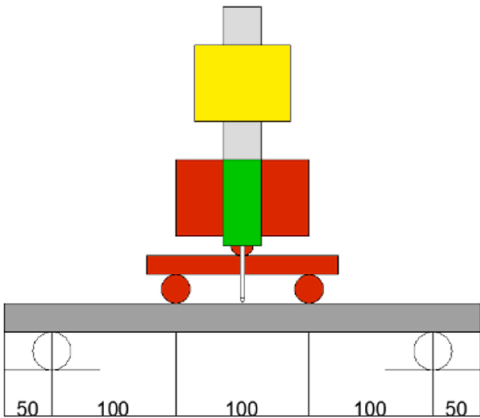


Fig. 3. Schematic diagram of four-point bending test.



Fig. 4. Loading device.

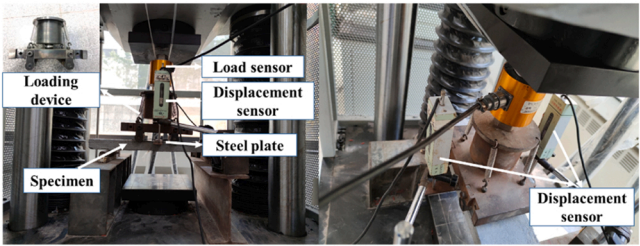


Fig. 5. Specimen loading.

solution and no external current is impressed, there is no change in the outer surface of the TRC permanent formwork and the stainless steel plate in the pool, and the water in the pool is relatively clear. As the test begins and current is impressed to the specimen, a small amount of bubbles are generated on the outer surface of the stainless steel plate. These bubbles adhere to the surface of the steel plate, and as the energization time of the specimen gradually increases, the bubbles gradually increase. This is because when stainless steel acts as the cathode in the ICCP system, the cathode will undergo a reduction reaction to produce some H_2 , and the chemical reaction is as shown in formula (1). In addition, with the prolongation of the energization time, a large number of bubbles were gradually generated on the outer surface of the TRC permanent formwork, which was attributed to the oxygen evolution and chloride evolution reaction at the anode material under the corrosion of chloride, which produced some O_2 and Cl_2 , and the chemical reaction was as shown in formulas (2–3). The water in the pool is yellowish brown and deepens with time, and the outer surface of the TRC permanent formwork has a darker yellowish brown color.



Table 8

Four-point bending test results of TRC permanent formwork.

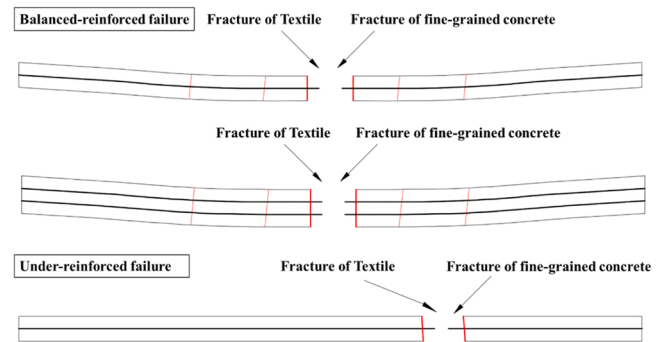
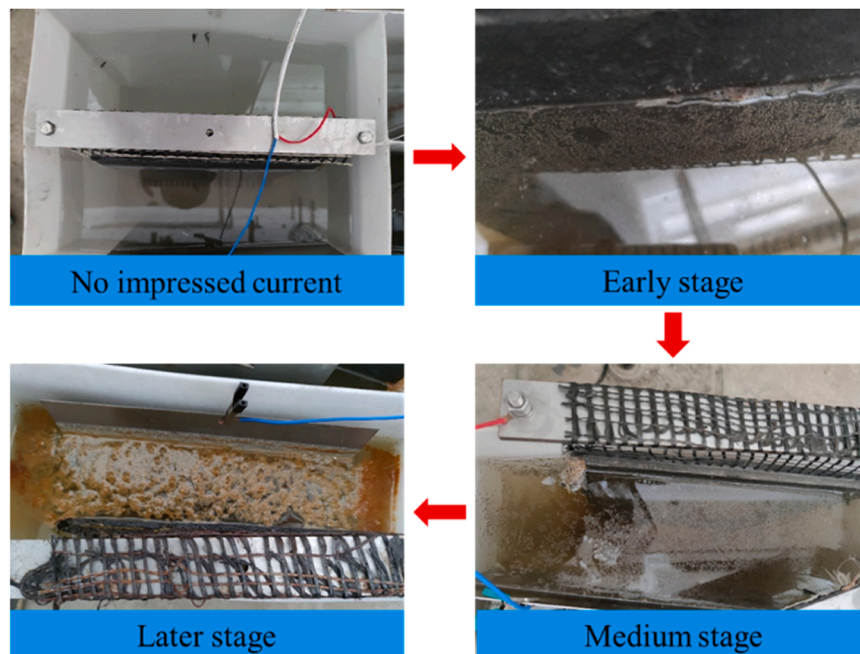
Specimen	First crack mid-span deflection /mm	First crack load /kN	Ultimate mid-span deflection /mm	Ultimate load /kN	Failure mode
B0	0.23	0.403	9.78	1.072	balanced-reinforced failure
B1	0.53	0.961	5.94	1.219	balanced-reinforced failure
B2	0.23	0.719	10.85	1.111	balanced-reinforced failure
B3	0.25	0.425	12.49	0.889	balanced-reinforced failure
B4	0.55	0.728	0.55	0.728	under-reinforced failure
B5	0.45	0.678	0.45	0.678	under-reinforced failure
B6	0.31	0.600	0.31	0.600	under-reinforced failure
B7	0.33	0.621	0.33	0.621	under-reinforced failure
B8	0.31	0.516	0.31	0.516	under-reinforced failure
B9	0.24	0.431	0.24	0.431	under-reinforced failure
B10	0.31	1.286	5.54	2.376	balanced-reinforced failure

3.3. Failure process and failure mode of the specimen

As shown in Fig. 7, the failure modes of different specimens are summarized and can be divided into two types of failure modes

(balanced-reinforced failure and under-reinforced failure). Regarding balanced-reinforced failure, there was no obvious change in the outer surface of the specimen in the early stage of loading; as the load further increased, multiple fine cracks appeared on the outer surface of the specimen; When the ultimate load is reached, the width of the main crack of the specimen suddenly expands, and the textile and fine-grained concrete in the direction of the force are fractured, and the specimen has obvious bending deformation when it is damaged, and the outer surface of the damaged specimen has fine and dense cracks. For under-reinforced failure, the development status of the specimen in the early stage of loading is similar to that of the specimen with balanced-reinforced failure, and its outer surface basically does not change; Continuing to increase the applied load, the specimen suddenly fractured when it reached the ultimate load. There was no obvious bending deformation when the specimen failed, and there was only one main crack on the outer surface.

Specimen B0–3 is in the chloride solution without impressed current, chloride solution on the mechanical properties of the textile has less impact, so that it can work with the fine-grained concrete under the action of the four-point bending load, and in the process of force to give full play to its own good tensile properties so that the specimen is damaged when the bending deformation is obvious, and the specimen has a number of cracks on the outer surface, belonging to the bending

**Fig. 7.** Four-point bending failure mode of TRC permanent formwork.**Fig. 6.** TRC permanent formwork state changes under chloride corrosion and impressed current.

damage. Specimen B4–9 is in a chloride solution with impressed current, and the tensile properties of the textile are severely degraded under the action of the impressed current, which does not allow it to effectively play its tensile role in the specimen, and makes it unable to work effectively with the fine-grained concrete when subjected to the action of four-point bending. Specimen premature destruction and the whole destruction process is more sudden, and did not show obvious bending deformation, its outer surface only a crack exists, in the crack can clearly see the fine-grained concrete and textile are fractured, belonging to the under-reinforced damage. There are two layers of textile in specimen B10. When an external current is impressed to the specimen, only one layer of the textile is directly exposed to an external current, while the other layer of textile still retains its original tensile properties because there is no external current. The specimen has obvious bending deformation when it fails, which is a balanced-reinforced failure.

3.4. Distribution of cracks in specimens

As shown in Fig. 8(a-d), it is the surface crack distribution of the specimen after four-point bending under different chloride concentrations during the non-energization process. It can be seen from the figure that the surface cracks of the damaged specimen B0–3 are mainly distributed in the pure bending section of the specimen, and the spacing between adjacent cracks is basically the same. In addition, when the specimen is in a chloride solution, the number of cracks on the outer surface of the specimen increases and the spacing decreases, but it is not obvious overall.

As shown in Fig. 8(e-g), when the specimen is exposed to a chloride solution in an impressed current environment, the crack development on the outer surface of the specimen changes significantly, and the number of cracks is significantly reduced. After the specimen is damaged, only one main crack can be seen on the outer surface of the specimen after damage. This is because under the action of four-point bending, the textile in the specimen mainly plays the role of delaying the damage of

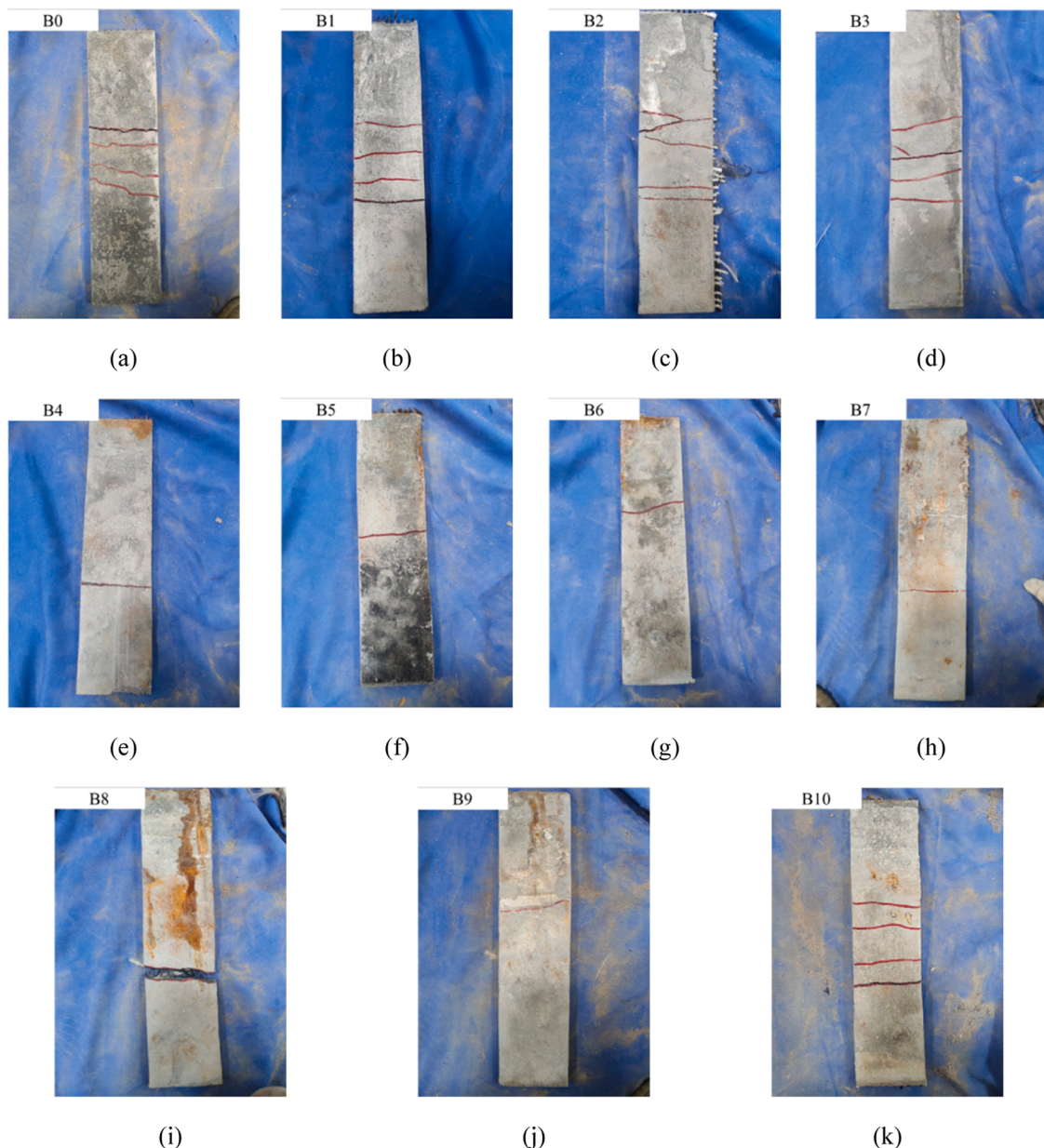


Fig. 8. Crack distribution of different TRC permanent formwork.

the specimen and improving the crack development state of the specimen. When an external current is impressed to the textile in the specimen, the impressed current will cause damage to the textile in the specimen, reducing the tensile properties of the textile, thus seriously affecting the crack distribution after the specimen is damaged.

As shown in Fig. 8(h-j), when the specimen is in a 2 % chloride concentration, as the quantity of electric charge density impressed to the specimen increases exponentially, the crack development pattern on the outer surface of the specimen after damage is basically the same as that of specimen B4–6, with only one crack on the surface of the specimen. This is because when the quantity of electric charge density is $0.35 \times 10^7 \text{C/m}^2$, the damage suffered by the textile has made it unable to provide effective tensile effect in the specimen. As the quantity of electric charge density increases, the damage becomes more severe and continues to reduce the mechanical properties of the specimen under four-point bending.

As shown in Fig. 8(k), even if the specimen is in an impressed current environment, the increase in the number of textile layers in the specimen can significantly improve the development of surface cracks when the specimen is damaged. The number of cracks on the surface of the specimen is significantly increased and the spacing is reduced, and the cracks develop into small and dense features.

3.5. Load-deflection curves of specimen under different environments

3.5.1. Effect of chloride concentration on load-deflection curve of specimen

Fig. 9(a) shows the load-deflection curve of the specimen with different chloride concentrations without current. The study found that the chloride concentration has a great influence on the load-deflection curve characteristics of the specimen. Compared with specimen B0, the first crack load of specimens B1–3 increased by 138 %, 78.4 %, and 5.5 % respectively. The ultimate load of specimens B1–2 increased by 13.7 % and 3.6 % respectively. The ultimate load of specimen B3 dropped by 17.1 %. It can be seen that when the chloride concentration is 2 %, the first crack load and ultimate load of the specimen increase to the greatest extent, and as the chloride concentration increases, the first crack load and ultimate load of the specimen continue to decrease. This is because, on the one hand, compared with the specimens not immersed in chloride solution, when the specimens are at a chloride concentration of 2 %, the crystals and corrosive substances produced by the fine-grained concrete in the specimens fill the pores in the concrete, enhancing the compactness of the internal structure of the concrete and improving the strength of the concrete in the compression zone; on the other hand, the chloride concentration has little effect on the tensile properties of the carbon fiber. This improves the initial crack and ultimate load of the specimen. However, as the chloride concentration increases, excessive corrosion substances are produced inside the concrete, destroying the dense structure of fine-grained concrete, gradually reducing the strength of the fine-grained concrete, and causing the initial crack and ultimate load to begin to decrease. Fig. 9(b) shows the four-point bending load-deflection curve of the specimen with different chloride concentrations and current. It can be seen from the figure that for the specimens in the electrified environment, as the chloride concentration increases, the first crack load and ultimate load of the specimens gradually decrease, but the two load values are not much different. This is because the impressed current damages the performance of the textile, affects its ability to work together with fine-grained concrete, and reduces the bending ability of the specimen.

3.5.2. Effect of impressed current on four-point bending load-deflection of specimen

Fig. 9(c), (d), and (e) are the load-deflection curves of the specimen without and with current at 2 %, 4 %, and 6 % chloride concentrations, respectively. It can be seen from the figure that under the same chloride concentration, the load-deflection curve characteristics of the specimen with impressed current undergo obvious changes. The load-deflection

curve of the specimen without impressed current has an obvious crack development stage, and multiple cracks occur on the outer surface of the specimen. However, the load-deflection curve of the specimen with impressed current has only two stages, that is, the specimen drops suddenly after the load reaches the peak value, and breaks suddenly, and there is only one crack on the outer surface of the specimen. In addition, when the chloride concentration is kept constant at 2 %, 4 %, and 6 %, the ultimate load of the specimen with impressed current is significantly reduced compared to the specimen without impressed current, which is reduced by 40.3 %, 39.0 %, and 32.5 % respectively. It can be seen that the bending performance of the impressed current specimen is severely degraded, which accelerates the damage and deterioration of the specimen. This is because the impressed current causes great damage and degradation to the textile inside the specimen, so that the textile cannot effectively exert its tensile properties when the specimen is subjected to four-point bending. When a large external load is applied, the fine-grained concrete and the textile breaks, causing the specimen to be damaged.

3.5.3. Effects of different quantity of electric charge densities on four-point bending load-deflection of specimen

Fig. 9(f) shows the load-deflection curve of the specimen under different quantity of electric charge densities ($0.35, 0.7, 1.4, 3.5 \times 10^7 \text{C/m}^2$). It can be seen from the figure that when the minimum quantity of electric charge density is $0.35 \times 10^7 \text{C/m}^2$, the load-deflection curves of the specimens after continuing to increase the quantity of electric charge density all show the same characteristics, that is, they include two development stages. In the pre-crack development stage of the specimen, the study found that as the quantity of electric charge density gradually increases, the first crack mid-span deflection of the specimen gradually decreases. In the failure stage of the specimen, compared with specimen B1, the ultimate loads of specimens B4, B7, B8, and B9 were reduced by 40.3 %, 49.1 %, 57.7 %, and 64.6 % respectively. It can be seen that as the quantity of electric charge density increases, the ultimate load of the specimen gradually decreases, indicating that the increase in quantity of electric charge density has an adverse effect on the bending performance of the specimen. In addition, by comparing the ultimate load reduction rates of specimens under different quantity of electric charge densities, it can be found that when the quantity of electric charge density increases to $0.35 \times 10^7 \text{C/m}^2$, the ultimate load reduction rate of specimens that continue to double the quantity of electric charge density becomes gradually smaller. This may be because the tensile properties of the textile inside the specimen are already very damaged under high quantity of electric charge density. At this time, continuing to increase the quantity of electric charge density will gradually reduce the negative impact on the four-point bending ultimate bearing capacity of the specimen.

3.5.4. Effects of different textile layers on four-point bending load-deflection of specimen

Fig. 9(g) shows the load-deflection curve of the specimen under different textile layers in an impressed current environment. It can be seen from the figure that the development characteristics of the load-deflection curves of B10 and B4 are obviously different. Even if the specimen B10 is in the process of impressed current, the load-deflection curve of specimen B10 with two layers of textile has an obvious crack development stage, and the ductility of the specimen is improved, and the specimen undergoes obvious balanced-reinforced failure. Compared with specimen B4, the ultimate load of specimen B10 increased by 226.4 %, and the ultimate mid span deflection increased by 907.3 %. This is because when there are two layers of textile in the specimen, when the current is impressed only to one layer, the other layer of textile can still effectively exert its tensile performance in the specimen, and work together with the damaged textile to improve the bending performance of the specimen. From this, it can be seen that textile plays a very important role in improving the bending performance of the

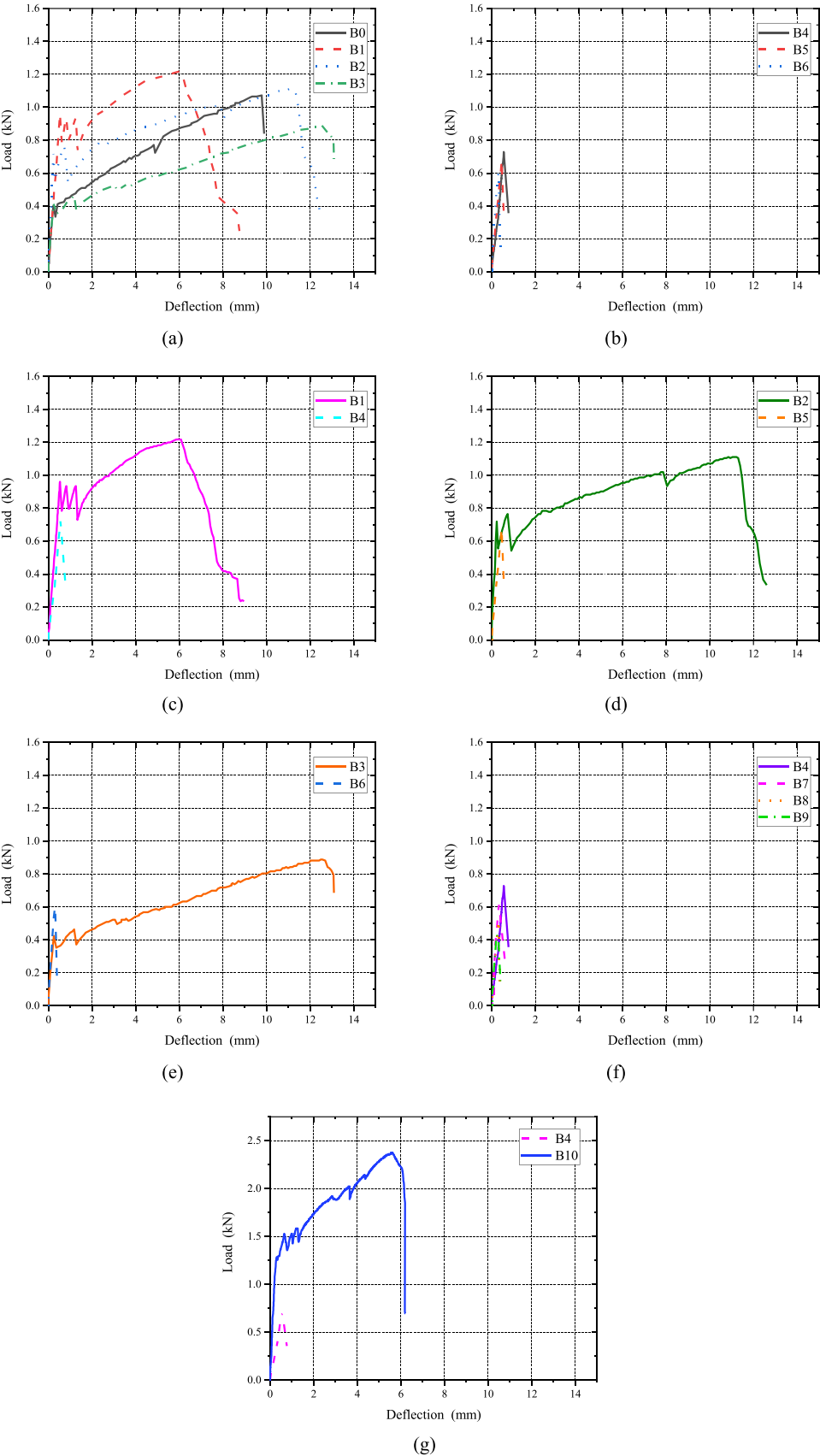


Fig. 9. Load-deflection curve of TRC permanent formwork under four-point bending.

specimen. When the bending performance of the specimen under impressed current is significantly reduced, multiple layers of textile can be arranged in the specimen to improve its bending performance.

3.6. Typical load-deflection curves of specimen under four-point bending

3.6.1. Typical 1 load-deflection curve

The load-deflection curve of specimen B0–3 can be summarized as typical 1. Because no external current is impressed to these four groups of specimens, and the chloride solution causes less damage to the textile. The fine-grained concrete and textile in the specimen work together to effectively change the development characteristics of the specimen during the failure process. As shown in Fig. 10 (a), it can be seen that the load-deflection curve of Typical 1 can be divided into four typical stages from the beginning to the end of loading: stage I (pre-cracking stage), stage II (multiple-cracking stage), stage III (crack widening stage) and stage IV (failure stage). Stage I refers to the stage before the first crack develops on the surface of the specimen. The applied bending load at this time is small, and the stress is mainly borne by the fine-grained concrete in the specimen, while the textile does not participate in the stress, and the load-deflection curve shows linear development characteristics. Stage II refers to the gradual formation of multiple small cracks on the surface of the specimen. When the applied load exceeds the tensile stress borne by the fine-grained concrete, the first crack appears on the surface of the specimen. The curve shows the first decrease in load as the deflection increases. As the bending load continues to increase, the cracked fine-grained concrete will transfer the tensile force to the textile, which will transfer the tensile force to other locations in the concrete. The powerful wave response at this stage is characterized by a jagged rise in load and deflection in the curve graph. Stage III means that the number of cracks on the surface of the specimen no longer increases, but the width of the cracks gradually expands as the bending force increases, showing a linear rise on the curve. Stage IV refers to the destruction stage of the specimen. As the external force gradually increases, a crack on the outer surface of the specimen expands sharply to form a main crack. The specimen was damaged at the main crack, and the textile was pulled out or broken. The curve shows that the load decreases sharply as the deflection increases.

3.6.2. Typical 2 load-deflection curve

The load-deflection curve of specimen B4–9 can be summarized as typical 2. This is because the specimens are in the chloride solution during the process of impressed current. The impressed current causes irreversible damage to the textile, which reduces its tensile properties. When the specimen is subjected to four-point bending load, the textile

cannot effectively transmit the external load, and the specimen does not have multiple cracks and cracks widening stages. As shown in Fig. 10 (b), it can be seen that the load-deflection curve of typical 2 can be summarized into two different development stages: stage I (pre-cracking stage) and stage II (failure stage). Stage I refers to the stage before cracks occur in the specimen. In the early stage of four-point bending loading, due to the small external force, the fine-grained concrete at the bottom of the specimen bears the bending force. There is no crack on the surface of the specimen until the first crack appears, and the curve shows that the load rises linearly as the deflection increases. Stage II refers to the failure stage of the specimen. When the bending force borne by the specimen increases to the ultimate load, the first crack appears on the surface of the specimen and failure occurs. This crack is both the first crack and the final crack of the specimen, that is, the first crack load is the ultimate load, which is shown in the curve as the load drops sharply after reaching the peak value. It can be seen that under chloride corrosion and impressed current, the load-deflection curve of the TRC permanent formwork with a single-layer textile shows the characteristics of Typical 2. However, for specimen B10, the load-deflection curve changed from typical 2 to typical 1, which was an obvious transformation. This is because there are two layers of textile in specimen B10. Even if the applied current causes damage to one layer of textile, the other layer of textile can still bear the bending effect better, thus significantly improving the bending performance of the TRC permanent formwork. This shows that increasing the number of textile layers is an effective method to improve the mechanical properties of TRC permanent formwork under the action of chloride corrosion and impressed current.

3.7. Effects of chloride corrosion and impressed current on the micromorphology of carbon fiber

As can be seen from Fig. 11 (a-b), the micromorphology of the outer surface of carbon fiber in specimens B0 and B3 is compared. It can be found that the gradual increase in chloride concentration has basically no obvious impact on the microscopic morphology of the outer surface of the carbon fiber monofilament, and the outer surface of the entire carbon fiber is still relatively smooth and complete. However, as the chloride concentration increases, the concrete corrosion products attached to the outer surface of the carbon fiber increase. This is mainly because as the chloride concentration increases, the corrosion products in the concrete increase, making the concrete constrained outside the carbon fiber more dense, thereby making it adhere to the carbon fiber.

As can be seen from Fig. 11 (c-f), the microscopic morphology of the outer surface of carbon fiber in specimens B4, B7, B8 and B9 is

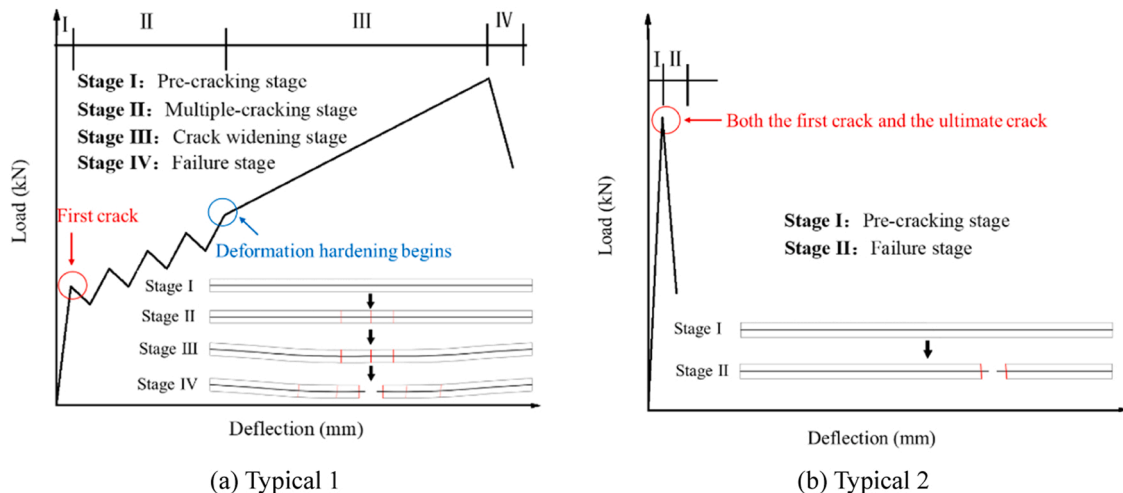


Fig. 10. Typical load-deflection curve under four-point bending.

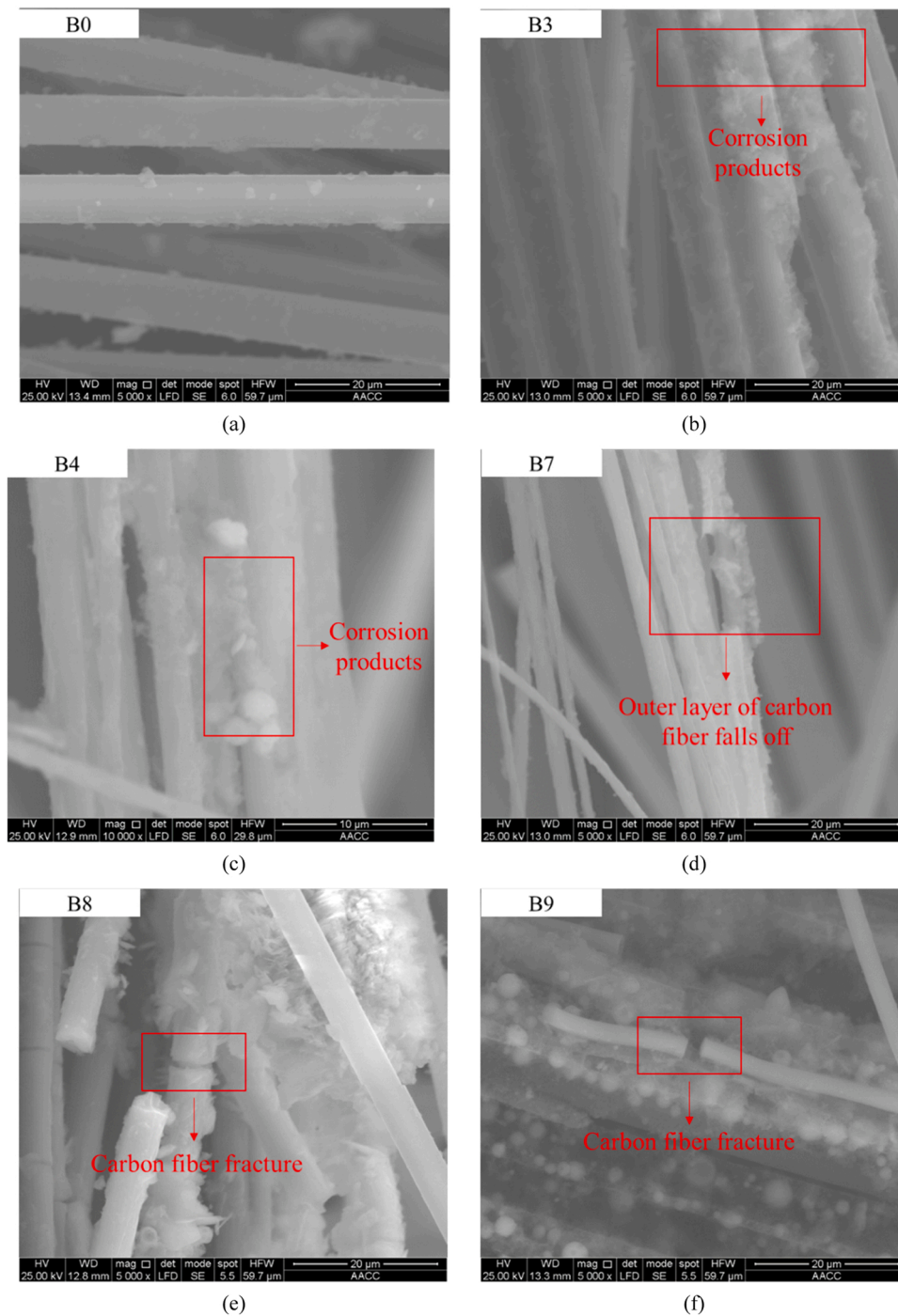


Fig. 11. Effects of chloride corrosion and impressed current on the micromorphology of carbon fiber.

compared. It can be found that due to the small quantity of electric charge density of specimen B4, the morphological characteristics of the carbon fiber have not shown obvious damage. However, as the quantity of electric charge density increases, the outer layer of the carbon fiber in specimen B7 begins to peel off. And the carbon fiber is still connected and has not completely broken. As the quantity of electric charge density increased to $1.4 (10^7 \text{C/m}^2)$, the carbon fiber in specimen B8 began to break. And when the quantity of electric charge density increases to $3.5 (10^7 \text{C/m}^2)$, the carbon fiber in specimen B9 has obvious fracture. This shows that when a current is supplied to the carbon fiber, the carbon fiber will be damaged as the quantity of electric charge density increases, resulting in the breakage of the carbon fiber.

4. Calculation model of four-point bending capacity of specimen under chloride corrosion and impressed current

Currently, there are few studies on the calculation model of the four-point bending capacity of TRC permanent formwork under impressed current. In order to make the TRC permanent formwork more effective in engineering practice, this section predicts the four-point bending capacity calculation model of the TRC permanent formwork based on experimental data and basic theory. This provides a theoretical basis for the design of TRC permanent formwork as anode material under impressed current cathodic protection method.

4.1. Theoretical analysis is based on the following basic assumptions

- (1) The section strain distribution obeys the plane section assumption
After the TRC permanent formwork is bent and deformed, its section remains flat, that is, the strain at any point in the section is proportional to the distance from the point to the neutral axis.
- (2) Ignore the influence of the tensile strength of fine-grained concrete on the cross-section bearing capacity
The TRC permanent formwork has fine-grained concrete near the neutral axis to bear the tensile force. However, its own tensile strength is low and its value is very small compared with the tensile force borne by the carbon fiber bundle, and the force is close to the neutral axis, so the bending moment it bears is ignored.
- (3) The interface bond between the textile and fine-grained concrete is reliable and there is no slip.
- (4) Ignore the action of fiber in non-tension directions.
- (5) Stress-strain relationship of fine-grained concrete under compression.

As shown in Fig. 12, since the strains in the concrete compression zone along the section height are not equal, the stress-strain relationship when fine-grained concrete is compressed uses the curve proposed in the current "Code for Design of Concrete Structures" (GB50010) [37]. The curve consists of a parabola and a horizontal straight line, and the theoretical formula is as follows:

$$\begin{cases} \sigma_c = f_c \left[1 - \left(1 - \frac{\varepsilon_c}{\varepsilon_0} \right)^n \right] & \varepsilon_c \leq \varepsilon_0 \\ \sigma_c = f_c & \varepsilon_0 < \varepsilon_c \leq \varepsilon_{cu} \end{cases} \quad (4)$$

$$n = 2 - \frac{1}{60} (f_{cu,k} - 50) \quad (5)$$

$$\varepsilon_0 = 0.002 + 0.5 (f_{cu,k} - 50) \times 10^{-5} \quad (6)$$

$$\varepsilon_{cu} = 0.0033 - (f_{cu,k} - 50) \times 10^{-5} \quad (7)$$

In the formula: σ_c —the compressive stress when the compressive strain of concrete becomes ε_c ;

f_c —Design value of concrete axial compressive strength;

ε_0 —the corresponding concrete compressive strain when the concrete compressive stress has just reached f_c . If the calculated value of ε_0 is less than 0.002, take 0.002;

ε_{cu} —the ultimate compressive strain of concrete in the normal section. When it is under non-uniform pressure, it is calculated according to the above formula. When the calculated ε_{cu} is greater

than 0.0033, take 0.0033;

$f_{cu,k}$ —Design value of compressive strength of concrete cube;

n —Coefficient, when the calculated n value is greater than 2.0, take 2.0.

- (6) Stress-strain relationship of carbon fiber bundles under tension

As shown in Fig. 13, carbon fiber bundles are anisotropic materials, and their stress-strain relationship develops linearly when subjected to tension. The theoretical formula are as follows:

$$\sigma_t = \varepsilon_t E_t \leq \sigma_{tu} \quad (8)$$

Where: σ_t is the stress of the carbon fiber bundle (MPa);

ε_t is the strain of the carbon fiber bundle;

σ_{tu} is the ultimate tensile strength of carbon fiber bundles (MPa);

E_t is the tensile elastic modulus of the carbon fiber bundle (GPa).

4.2. Calculation model of four-point bending capacity of specimen under normal environment

4.2.1. Cross-section stress analysis of specimen B0

Under the action of bending load of specimen B0, the textile in the tension area was broken, and the fine-grained concrete in the compression area was crushed. Therefore, when specimen B0 is damaged, the carbon fiber bundle strain is $\varepsilon_t = \varepsilon_{tu}$ and the concrete strain in the pressure zone is $\varepsilon_c^t = \varepsilon_{cu} = 0.0033$. At this time, the concrete stress distribution can be equivalent to a rectangular stress distribution. The principle of equivalence is that all forces in the two stress figure are equal and the positions of all forces application points remain unchanged. The cross-sectional calculation diagram is shown in Fig. 14.

The all forces F_c of concrete in the compression zone can be calculated as:

$$F_c = f_c b \xi_n h_t \left(1 - \frac{\varepsilon_0}{3\varepsilon_{cu}} \right) = \alpha_1 f_c \beta_1 \xi_n b h_t \quad (9)$$

Calculating the distance y_c between the all forces and the upper edge can be obtained:

$$y_c = \xi_n h_t \left(1 - \frac{\frac{1}{2} - \frac{\varepsilon_0^2}{12\varepsilon_c^2}}{1 - \frac{\varepsilon_0}{3\varepsilon_c}} \right) = 0.5 \beta_1 \xi_n h_t \quad (10)$$

From formulas (9–10), we get:

$$\alpha_1 = \frac{1}{\beta_1} \left(1 - \frac{\varepsilon_0}{3\varepsilon_{cu}} \right) \quad (11)$$

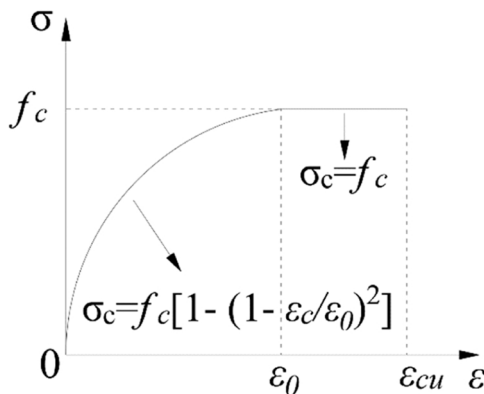


Fig. 12. Stress-strain relationship of fine-grained concrete under compression.

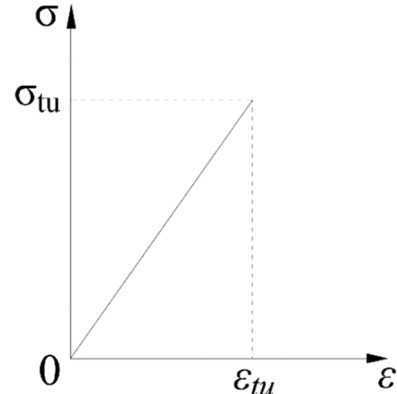


Fig. 13. Stress-strain relationship of carbon fiber bundles under tension.

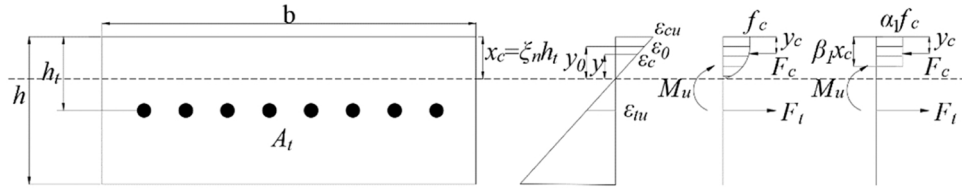


Fig. 14. Cross-sectional calculation diagram of specimen B0.

$$\beta_1 = \frac{1 - \frac{2\varepsilon_{c0}}{3\varepsilon_{cu}} + \frac{\varepsilon_{c0}^2}{6\varepsilon_{cu}^2}}{1 - \frac{\varepsilon_{c0}}{3\varepsilon_{cu}}} \quad (12)$$

In the formulas: α_1, β_1 is the non dimensional parameters for equivalent rectangular stress diagram.

ξ_n is the ratio of the height of the concrete compression zone to the effective height of the section.

b is the TRC permanent formwork section width;

h_t is the effective height of TRC permanent formwork section;

f_c is the axial compressive strength of fine-grained concrete.

4.2.2. Bearing capacity calculation model of specimen B0

Based on the above cross-section stress analysis diagram, two static formulas (13–14) can be obtained according to the static equilibrium state at the moment of destruction of the cross-section. In addition, according to the bending moment diagram under four-point bending, the relationship between the ultimate resistance bending moment M of the normal section and the four-point bending capacity F can be obtained:

$$\alpha_1 f_c \beta_1 \xi_n b h_t = A_t \sigma_{tu} \quad (13)$$

$$M_u = \alpha_1 f_c \beta_1 \xi_n b h_t^2 (1 - 0.5 \beta_1 \xi_n) \quad (14)$$

$$M = \frac{F}{6} l \quad (15)$$

In the formulas: A_t is the longitudinal tensile cross-sectional area of the carbon fiber bundle;

σ_{tu} is the ultimate tensile strength of carbon fiber bundles;

M_u is the ultimate bending moment resistance of the normal section of the TRC permanent formwork.

l is the span of the two bottom supports when the TRC permanent formwork is bent at four points.

4.2.3. Verification of bearing capacity calculation model

From the above formulas (13–15), according to $f_c = 31.13\text{MPa}$, $b = 100\text{mm}$, $\sigma_{tu} = 3600\text{MPa}$, $h_t = 6\text{mm}$, $A_t = 3.6\text{mm}^2$. The results are shown in Table 9. It can be seen that the theoretical values calculated by the above formula are basically consistent with the experimental values.

4.3. Calculation model of four-point bending capacity of specimen under chloride corrosion and impressed current

4.3.1. Influence coefficient of concrete strength under different chloride concentrations

Through the above research, it was found that changes in external chloride concentration have little impact on the mechanical properties of the textile in the specimen, but mainly have a certain impact on the compressive strength of the fine-grained concrete in the specimen. Therefore, this paper introduces the influence coefficient γ of concrete

strength when calculating the bearing capacity of specimen under different chloride concentrations. In addition, when specimen B1–3 fails, the carbon fiber bundles in the tension area are broken and the fine-grained concrete in the compression area is crushed. Its cross-sectional stress diagram is consistent with that of specimen B0. The obtained formula is as follows:

$$\alpha_1 \gamma f_c b \beta_1 \xi_n h_t = A_t \sigma_{tu} \quad (16)$$

$$M_u = \alpha_1 \gamma f_c b \beta_1 \xi_n h_t \left(h_t - \frac{\beta_1 \xi_n h_t}{2} \right) \quad (17)$$

As shown in Fig. 15, according to formulas (16–17) and the data obtained from the experiments in this paper, the relationship between concrete strength parameters and chloride concentration is fitted:

$$\gamma = 2.06889 - 0.2051Cl \quad (18)$$

In the formula: Cl is the chloride concentration, and the range is 2–6 (%).

4.3.2. Damage coefficient of carbon fiber bundle under impressed current

When specimen B10 was damaged, both layers of textile in the tension area were broken, and the fine-grained concrete in the compression area was crushed. The failure mode showed balanced-reinforced failure. The cross-sectional stress is shown in Fig. 16. In addition, research shows that the main reason why the quantity of electric charge density reduces the bending performance of the specimen is that the carbon fiber bundle in the specimen is damaged under the action of current, and the impact on the strength of fine-grained concrete can be ignored. In specimen B10, there is only one textile with an impressed current. Therefore, this paper considers introducing the damage coefficient η of carbon fiber bundle to characterize the impact of external quantity of electric charge density on the four-point bending capacity of the specimen. The calculation model can be obtained as:

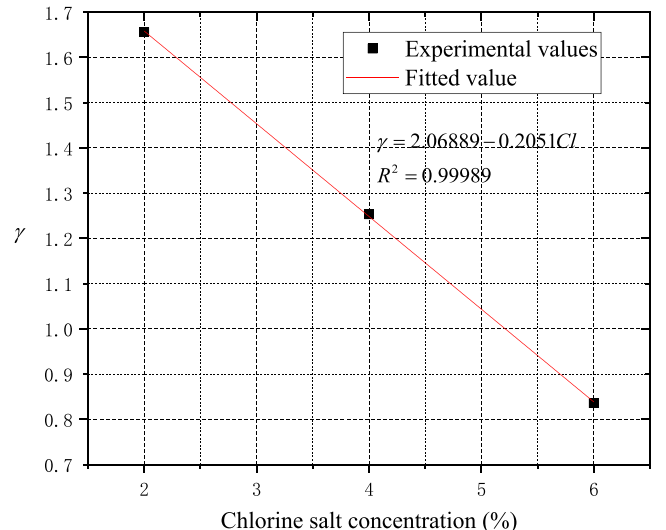


Fig. 15. Influence coefficient of concrete strength.

Table 9

Comparison of theoretical and experimental values of specimen B0.

Specimen	Theoretical value/N	Experimental values /N	Theoretical value / Experimental values
B0	1016	1072	0.947

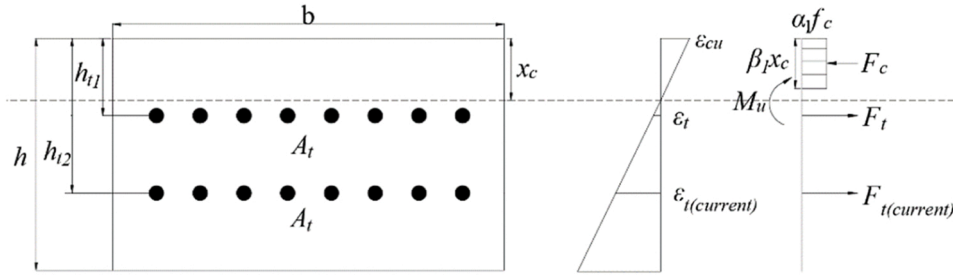


Fig. 16. Cross-sectional calculation diagram of specimen B10.

$$\alpha_1 \gamma f_c b \beta_1 x_c = \sigma_{tu} A_t + \sigma_{t(current)u} A_t \quad (19)$$

$$\sigma_{t(current)u} = \eta \sigma_{tu} \quad (20)$$

$$M_u = \sigma_{tu} A_t \left(h_{t1} - \frac{\beta_1 x_c}{2} \right) + \sigma_{t(current)u} A_t \left(h_{t2} - \frac{\beta_1 x_c}{2} \right) \quad (21)$$

In the formula: $\sigma_{t(current)u}$ is the ultimate tensile strength of carbon fiber bundles under impressed current;

h_{t1} is the position of the first layer of textile (6 mm);

h_{t2} is the position of the second layer of textile (12 mm).

According to formula (19–21), and the B10 ultimate load data obtained in the test of this paper, when the current density is $0.35 (10^7 \text{C}/\text{m}^2)$, the damage coefficient of carbon fiber bundle $\eta = 0.5106$.

4.3.3. Bearing capacity calculation model

(1) Cross-section stress analysis

According to the test results, when the specimen B4–9 is damaged by chloride corrosion and impressed current, the textile in the tension area is broken, and the fine-grained concrete in the pressure area is not crushed, which is a under-reinforced failure. At this time, the fine-grained concrete in the compression zone is in the elastic-plastic stage ($\epsilon_{c0} \leq \epsilon_c^t < \epsilon_{cu}$). The cross-sectional stress analysis is shown in Fig. 17.

It can be seen from formula (4) that the concrete stress-strain relationship can be obtained:

$$\begin{cases} \sigma_c = f_c \left[1 - \left(1 - \frac{\epsilon_c}{\epsilon_0} \right)^2 \right] & \epsilon_c \leq \epsilon_0 \\ \sigma_c = f_c & \epsilon_0 < \epsilon_c \leq \epsilon_{cu} \end{cases} \quad (22)$$

In the formula: $\epsilon_0 = 0.002$, $\epsilon_{cu} = 0.0033$.

According to the geometric relationship of strain distribution, get:

$$\frac{\epsilon_c^t}{\xi_n h_t} = \frac{\epsilon_t}{(1 - \xi_n) h_t} = \frac{\epsilon_c}{y} = \frac{\epsilon_0}{y_0} \quad (23)$$

The all forces F_c of concrete in the compression zone can be calculated as:

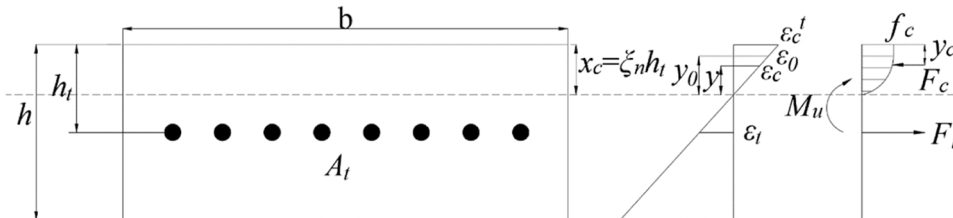


Fig. 17. Cross-sectional calculation diagram of specimen B4–9.

$$F_c = b \int_0^{y_0} f_c \left[1 - \left(1 - \frac{\epsilon_c}{\epsilon_0} \right)^2 \right] dy + b \int_{y_0}^{\xi_n h_t} f_c dy = f_c b \xi_n h_t \left(1 - \frac{\epsilon_0}{3 \epsilon_c^t} \right) \quad (24)$$

The all forces F_c in the concrete compression zone is taken as the moment about the neutral axis, get:

$$F_c \left(\xi_n h_t - y_c \right) = b \int_0^{y_0} f_c \left[1 - \left(1 - \frac{\epsilon_c}{\epsilon_0} \right)^2 \right] y dy + b \int_{y_0}^{\xi_n h_t} f_c y dy \quad (25)$$

The distance y_c from the all forces F_c in the concrete compression zone to the edge of the concrete is as shown in the following formula:

$$\begin{aligned} y_c &= \xi_n h_t - \frac{b \int_0^{y_0} f_c \left[1 - \left(1 - \frac{\epsilon_c}{\epsilon_0} \right)^2 \right] y dy + b \int_{y_0}^{\xi_n h_t} f_c y dy}{b \int_0^{y_0} f_c \left[1 - \left(1 - \frac{\epsilon_c}{\epsilon_0} \right)^2 \right] dy + b \int_{y_0}^{\xi_n h_t} f_c dy} \\ &= \xi_n h_t \left(1 - \frac{\frac{1}{2} - \frac{\epsilon_0^2}{12 \epsilon_c^t}}{1 - \frac{\epsilon_0}{3 \epsilon_c^t}} \right) \end{aligned} \quad (26)$$

(2) Bearing capacity calculation model

Based on the above cross-sectional stress analysis diagram, by introducing the concrete influence coefficient and the damage coefficient of carbon fiber bundle, it can be obtained.

$$\gamma f_c b \xi_n h_t \left(1 - \frac{\epsilon_0}{3 \epsilon_c^t} \right) = \eta \sigma_t A_t \quad (27)$$

$$\begin{aligned}
 M_u &= \gamma f_c b \xi_n h_t^2 \left(1 - \frac{\varepsilon_0}{3\varepsilon_c^t} \right) \left[1 - \xi_n \left(1 - \frac{\frac{1}{2} - \frac{\varepsilon_0^2}{12\varepsilon_c^2}}{1 - \frac{\varepsilon_0}{3\varepsilon_c^t}} \right) \right] \\
 &= \eta \sigma_t A_t h_t \left[1 - \xi_n \left(1 - \frac{\frac{1}{2} - \frac{\varepsilon_0^2}{12\varepsilon_c^2}}{1 - \frac{\varepsilon_0}{3\varepsilon_c^t}} \right) \right] \quad (28)
 \end{aligned}$$

(3) Verification of bearing capacity calculation model

Specimen B4–6 is in an environment with chloride concentration of 2 % and quantity of electric charge densities of 0.35 (10⁷C/m²). The theoretical bearing capacity of specimen B4–6 can be calculated according to formula (27–28). Comparing the theoretical value of bearing capacity with the experimental value, it can be found that the two can match well. The results are shown in Table 10. In addition, for the calculation model after the introduction of influence coefficients for concrete strength and damage coefficients for carbon fiber bundles, it was found through analysis that the carbon fiber damage coefficient has a greater impact on the calculation model. However, since the degradation damage coefficient of the carbon fiber under high quantity of electric charge density is not yet known, further experiments and in-depth analysis are needed to evaluate its impact on the calculation model.

5. Conclusion

This paper studies the bending and durability performance of TRC permanent formwork under chloride corrosion and impressed current, using a combination of experiments and theory to lay the foundation for the promotion and application of TRC permanent formwork in the field of ICCP technology. The main conclusions are as follows:

- (1) When the specimen is in a conventional environment and a single chloride corrosion, the failure mode of the specimen under the four point bending test is balanced-reinforced failure; When the specimen is subjected to chloride corrosion and impressed current, the failure mode of the specimen undergoes a transition, resulting in under-reinforced failure; Increasing the number of textile layers in the specimen can shift the failure mode from under-reinforced failure to balanced-reinforced failure.
- (2) Compared with specimens in a conventional environment and a single chloride corrosion, when the specimens are subjected to chloride corrosion and impressed current, the morphology of surface cracks on the specimens undergoes a significant change, and the number of cracks decreases significantly; Increasing the number of textile layers can significantly improve the distribution of surface cracks during specimen failure under chloride corrosion and impressed current.
- (3) Compared with specimens in a conventional environment and a single chloride corrosion, the bending performance of the specimens deteriorates more severely when subjected to chloride corrosion and impressed current. This is because the current causes significant damage and degradation to the carbon fiber inside the specimens. In addition, the ultimate load of the specimen gradually decreases as the electric charge density increases.
- (4) The typical load-deflection curve of the specimen is mainly divided into the following two categories. Typical 1, the load-deflection of the specimen can be divided into four typical stages: pre-cracking stage, multiple-cracking stage, crack widening stage and failure stage. Typical 2, the load-deflection of the specimen can be divided into pre-cracking stage and failure stage.

Table 10

Comparison between theoretical values and experimental values of specimen B4–6.

Specimen	theoretical values /N	experimental values /N	theoretical values / experimental values
B4	692	728	0.951
B5	664	678	0.979
B6	611.2	600	1.019

- (5) By analyzing the microstructure of the outer surface of carbon fiber in different specimens, it can be found that current will cause damage to the carbon fiber, and with the increase of electric charge density, the surface of the carbon fiber will experience more obvious fracture damage.
- (6) This paper is based on basic assumptions, combined with test data and the cross-sectional stress analysis diagram when the specimen is damaged. A calculation model for the four-point bending capacity of TRC permanent formwork under different chloride concentrations and quantity of electric charge densities under chloride corrosion and impressed current was established.

CRedit authorship contribution statement

Farhad Aslani: Resources, Formal analysis. **Shiping Yin:** Supervision, Funding acquisition, Conceptualization. **Tianhao Qin:** Writing – review & editing. **Shichang Li:** Writing – original draft, Methodology, Data curation, Conceptualization.

Declaration of Interest Statement

We declare that there have not any commercial or associative interests that represent a conflict of interest in connection with this work submitted.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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